

APPLE (AAPL) HISTORICAL STOCK DATA ANALYSIS REPORT

Dataset Overview

This analysis used the Apple Inc. (AAPL) Historical Stock Dataset from Yahoo Finance covering June 09, 2021 to June 09, 2026.

The dataset contains daily stock information, including Date, Open, High, Low, Close, and Trading Volume.

The objective was to analyse Apple's stock performance, identify trends, create useful features, and generate insights for investment decisions.

Data Transformation

The following steps were performed:

- Handled missing values and multi-valued column headers
- Converted Date to datetime format
- Extracted Year, Month, and Day
- Rounded price columns to 2 decimal places
- Sorted and formatted data
- Calculated monthly averages
- Prepared data for time-series analysis

Features Created

- Daily Price Change
- Daily Price Change %
- Daily Return
- 7-Day Moving Average (MA7)
- 30-Day Moving Average (MA30)
- Monthly Average Closing Price
- Volatility

Key Insights and Recommendation

Closing Price Trend: This chart was created to understand the overall movement of Apple's stock price over the five-year period.

Insight: Apple's stock price showed a strong upward trend despite temporary declines in 2022–2023 and 2025. The stock recovered and reached new highs in 2026.

Recommendation: Long-term investors may benefit more from a buy-and-hold strategy than short-term trading.

Trading Volume Trend: This chart was used to examine investor participation and trading activity over time.

Insight: Daily trading volume generally ranged between 50M and 100M shares, with occasional spikes above 200M. Despite fluctuating volume, the stock price continued to rise, indicating strong investor confidence.

Recommendation: Investigate unusual volume spikes as they may be linked to important company or market events.

Moving Average Analysis: This chart was used to identify the short-term and long-term direction of Apple's stock price.

Insight: Both the 7-day and 30-day moving averages trended upward, confirming long-term growth. Temporary bearish crossovers occurred but were followed by recovery.

Recommendation: Focus on the long-term trend rather than reacting to short-term price changes.

Daily Return Analysis: This chart was used to measure the daily gain or loss in Apple's stock price.

Insight: Daily returns fluctuated around zero, with most gains and losses being relatively small. This indicates reasonable price stability.

Recommendation: Daily returns should be analysed alongside volume and market news before making investment decisions.

Volatility Analysis: This chart was used to measure how stable or risky Apple's stock price was over time.

Insight: Volatility remained moderate for most of the period, with a notable spike around 2025 indicating increased uncertainty.

Recommendation: Monitor periods of high volatility, but evaluate them within the broader market trend.

Conclusion

Apple maintained strong long-term growth between 2021 and 2026. Although periods of price decline, volatility spikes, and fluctuating trading volume occurred, the overall trend remained positive. The analysis confirms strong investor confidence, stable performance, and continued long-term growth potential.